

Polynomials (Part II): Long Division (Homework)

First write
the dividend
and/or
the divisor
in descending
order of
powers.

Perform the following division

① $(x^3 + 5x^2 + 2x - 8) \div (x + 4)$

② $(3x^4 + 4x^3 - 9x^2 - 18x - 8) \div (3x^2 - 2x - 8)$

③ $(-x^4 - 6x^3 - 9x^2 + 4x + 12) \div (x - 1)$

④ $(x^4 - x^3 + 5x^2 - 2x + 7) \div (4x^2 + 3 + x^3)$

⑤ $(-8x^2 - 4x^3 - x + 10) \div (x - 2)$

⑥ $(2x^3 + 5x^2 + x - 1) \div (x^2 + 1)$

⑦ $(-6x^5 - 5x^3 - 2x^4 + x + 8x^2 + 2) \div (x + 2x^3)$

⑧ $(x^5 - 2x^3 + 8) \div (x + 2)$

⑨ $(3x^4 + 9x - 17) \div (x^2 + 3x - 10)$

⑩ $(9x^4 + 2x^3 + 5x + 15) \div (x^2 + 1)$

⑪ $(2x^5 - 2x + 3x^4 - 3) \div (3 + 2x)$

Answers

① $x^2 + x - 2$

② $x^2 + 2x + 1$

③ $-x^3 - 7x^2 - 16x - 12$

④ $5x^2 - 21x + 70 + \frac{-219x - 203}{x^2 + 4x + 3}$ or $5x^2 - 21x + 70 - \frac{219x + 203}{x^2 + 4x + 3}$

⑤ $4x^2 - 1 + \frac{8}{x-2}$

⑥ $2x + 9 + \frac{-x - 10}{x^2 + 1}$ or $2x + 9 - \frac{x + 10}{x^2 + 1}$

⑦ $-3x^2 - x - 1 + \frac{9x^2 + 2x + 2}{2x^2 + x}$

⑧ $x^4 - 2x^3 + 2x^2 - 4x + 8 - \frac{8}{x+2}$

⑨ $3x^2 - 9x + 57 + \frac{-252x + 553}{x^2 + 3x - 10}$

⑩ $8x^2 + 2x - 8 + \frac{3x + 23}{x^2 + 1}$

⑪ $x^4 - 1$